

Homework Assignment 2

Show all your computations and justify your answers.

1. (30 points) Consider the function $f(x) = \sqrt{\frac{1}{4x^3-8}}$.

- (a) Show that f is one-to-one.
- (b) Compute $f^{-1}(x)$ and find its domain and range.
- (c) Compute $\lim_{x \rightarrow \frac{1}{10}} f^{-1}(x)$.

2. (21 points) Calculate the following limits.

- (a) $\lim_{x \rightarrow \sqrt{2}} \sqrt{\frac{60 - 22x^2}{2\sqrt{2} - x}}$.
- (b) $\lim_{x \rightarrow -2} \frac{x^3 + x^2 - 2x}{x + 2}$.
- (c) $\lim_{x \rightarrow 0} \frac{7x^2}{|x|}$.

3. (20 points) Given $\varepsilon = \frac{1}{5}$, find δ such that:

- (a) $|x - 1| < \delta$ implies $|x^2 + x - 2| < \varepsilon$.
- (b) $|x - 2| < \delta$ implies $|\frac{1}{x} - \frac{1}{2}| < \varepsilon$.

4. (14 points) Using ε and δ definition of limits prove that $\lim_{x \rightarrow 2} \frac{1}{x} = \frac{1}{2}$.

5. (15 points) Prove that $\lim_{x \rightarrow 0} \left(\sin^2 \left(\frac{1}{x^2} \right) \sin^2(x^2) \right) = 0$.